Data Sources

Today’s Agenda

- Causal Inference: Adjusting for Observables
- Clarification on ‘economics’ requirement
- Finding Data
- Evaluating Data
- Working time
  - Download a dataset for your project
  - Verify that it has the right info
  - Check the size
  - Run an initial analysis
  - Record all this in Canvas survey
- Peer Reviews

Adjusting for Observables

- Subclassification (Cochran 1968)
- Nearest Neighbor Matching (Abadie and Imbens 2006, 2008
- Propensity Score (Rosenbaum and Rubin 1973)
- Multivariate Regression (Control Variables)

Identification under independence

- Recall randomization implies independence:
$$(Y^0, Y^1) \perp D$$
- Which means
$$E[Y|D = 1] - E[Y|D = 0] = E[Y^1 - Y^0] = ATE$$
- And also ATE=ATT because
$$E[Y^1 - Y^0] = E[Y^1 - Y^0|D = 1]$$

Conditional Independence

- Identification assumptions

1. $(Y^0, Y^1) \perp D \mid X$
2. $0 < \Pr(D = 1|X) < 1$ with probability 1 (common support)

- With those two assumptions, we can calculate the ATE
- If we just want the ATT, we can relax to

1. $Y^0 \perp D \mid X$
2. $\Pr(D = 1|X) < 1$ with $\Pr(D = 1) > 0$

Subclassification Estimator

- Basic idea:

$$\delta_{ATE} = \int E[Y|X, D = 1] - E[Y|X, D = 0]dPr(X)$$

- Intuition: compare treatment and control for subgroups weighted by the probability that individuals in that subgroup are treated.

Example: Israeli COVID Data

- Nearly 60% of new patients hospitalized for COVID as of August 15, 2021 are vaccinated.
- Does this mean vaccines are ineffective?

Israeli COVID Data

Several problems:

- Vaccination rates are high. (Base rate fallacy)
- Older people are especially likely to be vaccinated, younger people less so. (Selection)

Adjusting for Vaccination Rates

- Suppose vaccines were randomly assigned, then we could calculate a difference in means:

Age	Population (%)		Severe cases		Efficacy
	Not Vax %	Fully Vax %	Not Vax per 100k	Fully Vax per 100k	vs. severe disease
All ages	1,302,912 18.2%	5,634,634 78.7%	214 16.4	301 5.3	**67.5%**

Selection Bias

- But selection bias still remains! What to do?

Subclassification

- Let’s look at two big age groups:

Age	Population (%)		Severe cases		Efficacy
	Not Vax %	Fully Vax %	Not Vax per 100k	Fully Vax per 100k	vs. severe disease
All ages	1,302,912 18.2%	5,634,634 78.7%	214 16.4	301 5.3	**67.5%**
<50	1,116,834 23.3%	3,501,118 73.0%	43 3.9	11 0.3	**91.8%**
>50	186,078 7.9%	2,133,516 90.4%	171 91.9	290 13.6	**85.2%**

More subclassification

- Breaking it down even further

Age	Population (%)		Severe cases/100k		Severe Case Risk	Efficacy
	% Not Vax	% Fully Vax	Not Vax	Fully Vax	Ratio w/ 30-39 UnVax	vs. severe disease
12-15	62.1%	29.9%	0.30	0.00	1/20x	**100%**
16-19	21.9%	73.5%	1.60	0.00	1/4x	**100%**
20-29	20.5%	76.2%	1.50	0.00	1/4x	**100%**
30-39	16.2%	80.9%	6.20	0.20	1	**96.8%**
40-49	13.2%	84.4%	16.50	1.00	2.7x	**93.9%**
50-59	10.0%	88.0%	40.20	2.90	6.5x	**92.8%**
60-69	8.8%	89.8%	76.60	8.70	12.4x	**88.7%**
70-79	4.2%	94.6%	190.10	19.80	30.7x	**89.6%**
80-89	5.6%	92.6%	252.30	47.90	40.7x	**81.1%**
90+	6.1%	90.5%	510.9	38.60	82.4x	**92.4%**

Other methods

- Subclassification is often not feasible with lots of covariates.
- Options:
  - Control for X. Works well if we believe that a linear model is accurate.
  - Nearest neighbor matching: impute the missing potential outcome of each treatment unit by using the observed outcome from that outcomes nearest neighbor j (or an average of several neighbors)
    - Problem: how to define ‘nearness’? Several options in practice.
  - Propensity score matching calculates a probability of treatment for each unit, and we compare with units with similar estimates probabilities of treatment. Adjust for propensity score instead of X.
- Remember: NONE of these methods address selection on unobservables, which economists frequently worry about.

Important Clarification

- If you’re relying on econometrics for the ‘economics’ element of your project, make sure you include causal analysis.
- You don’t need an A/B test for causality – other options such as instrumental variables, regression discontinuity, matching, may work.
- However, simple OLS regression is unlikely to cut it UNLESS you have randomly assigned treatment.

Finding Good Data

- What are we looking for?
  - A reliable source: government, trusted company, etc.
  - A large sample size.
  - Ideally, we want it to be *representative* of something.
- Common pitfalls
  - The data says it has what we want, but the values are missing.
  - The documentation is awful or nonexistent.
  - The dataset is not large enough (especially for ML-type approaches).

Evaluating Data

- Download the dataset ASAP!
  - Descriptions can be misleading.
  - Unexpected challenges ALWAYS arise.
- Are the data you’re interested in:
  - Appropriately linked to other data? Or linkable?
  - Non-missing?
- Today
  - Start w/ a list of datasets.
  - Download one.
  - Make some scatterplots/ histograms. Seem sensible?
  - Share a simple interesting graphic or table.

Bureau of Labor Statistics

- Consumer price indices (inflation).
- Unemployment
- Wage data
- Productivity
- More specific price indices
- More!

Living Standards Measurement Surveys

- Detailed household data from around the world
  - Food consumption
  - Income
  - Wealth
  - Crop production
  - Women empowerment
  - More!
- https://www.worldbank.org/en/programs/lsms

Bloomberg

- Bloomberg has very detailed financial data on most anything you could imagine.
- USF has a Bloomberg Lab – let me know if you'd be interested in gaining access and I'll try to find a way.
- For today, focus on alternatives, since this won't be easy to find. Google/ Yahoo Finance could be good places to start.